

Which encoding strategy (composite or dummy) provides better predictive performance?

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Composite (ordinal) encoding provides better predictive performance than dummy encoding. Across 30 repetitions of 10-fold cross-validation, the composite model achieved a mean R^2 of **0.060** (95% CI [0.047, 0.072]) compared to **0.003** (95% CI [-0.011, 0.017]) for the dummy model, and it also produced lower RMSE (0.870 vs. 0.894) and lower MAE (0.716 vs. 0.725). The advantage held in **every one of the 30 repetitions** for both R^2 and RMSE, and in 28 of 30 for MAE.

Composite encoding represents each transportation question with a single ordered variable, whereas dummy encoding creates a separate indicator variable for every response category. Starting from the merged dataset, I scored communication apprehension from the 12 PRCA items after reverse-scoring the positively worded ones, built both encodings from the same 243 complete respondents, and fit identical linear regression models that differed only in encoding. Both models were evaluated on the **same** repeated 10-fold cross-validation splits (30 repetitions), so every difference in R^2 , RMSE, and MAE comes from a paired comparison on identical data; means were compared with bootstrap confidence intervals across the 30 repetitions.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

repeat_results = pd.read_csv("../data/results/encoding_cv_repeat_results.csv")

plot_data = repeat_results.rename(
    columns={
```

```

        "r2_composite": "Composite",
        "r2_dummy": "Dummy"
    }
).melt(
    id_vars="repeat",
    value_vars=["Composite", "Dummy"],
    var_name="Encoding Strategy",
    value_name="Cross-validated R2"
)

plt.figure(figsize=(6,4))

sns.boxplot(
    data=plot_data,
    x="Encoding Strategy",
    y="Cross-validated R2"
)

sns.stripplot(
    data=plot_data,
    x="Encoding Strategy",
    y="Cross-validated R2",
    alpha=.55
)

plt.tight_layout()
plt.show()

```

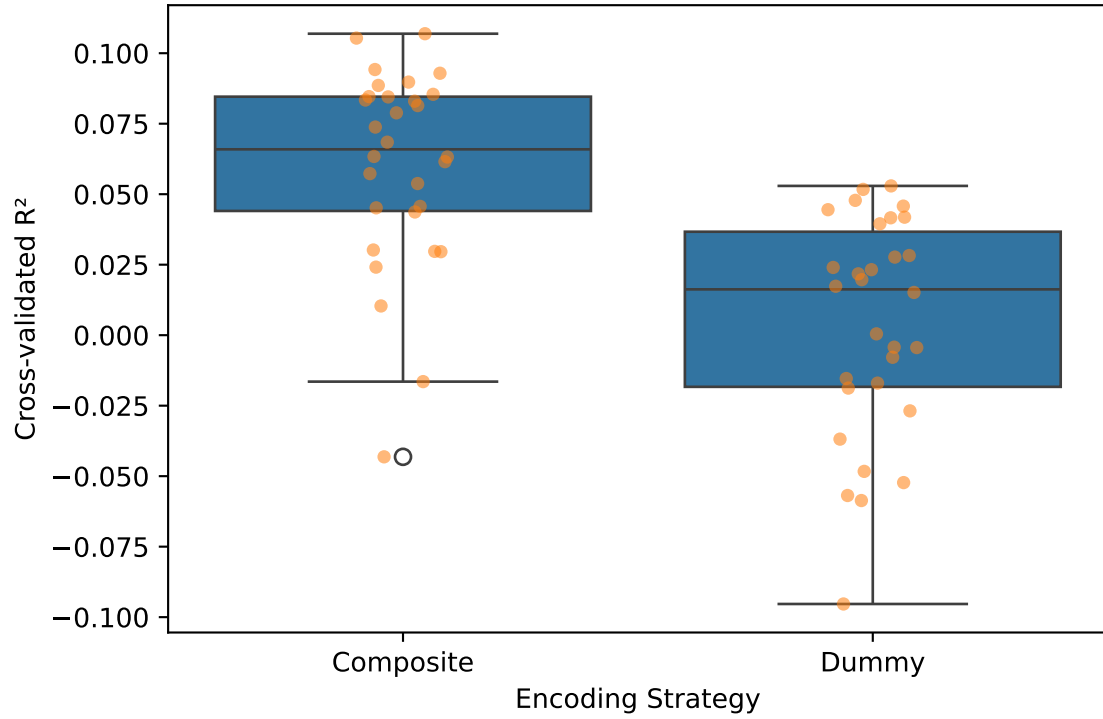


Figure 1: Repeated cross-validated R^2 values for the composite and dummy encoding strategies.

Figure 1 shows the performance advantage was consistent across repeated cross-validation rather than being driven by a single favorable train-test split: the paired composite-minus-dummy difference in mean cross-validated R^2 was **+0.057** (95% CI [0.051, 0.063]), with composite also producing consistently lower prediction error.